

Assignment 2: Choose Your Own Analysis?

A data-driven analysis using MADOC to examine user discussions, sentiment, and trends within Reddit gaming community

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1. Introduction

Background

Gaming communities have become a main source of online social media interaction with platforms like Reddit getting millions of user interactions daily about games. The r/gaming subreddit is the largest gaming community within Reddit that serves as a central location where users share their opinions about various game topics, ranging from game mechanics, technical issues to broader debates about the gaming culture.

Problem Formulation

Within the online gaming communities, toxicity has always been a persistent theme over the years. Toxic behaviour includes aggressive complaints or personal attacks about games and fellow users. These behaviours lead to negative consequences where the overall community morale is negatively impacted, especially when more users adopt such behaviours. Therefore, the team wanted to understand how toxicity in online gaming spaces manifests within distinct discussion communities where a group of users frequently interact with each other through replies/comments.

Research Question

This led to the development of our research question:

“Which gaming communities exhibit the most negative sentiment, and what roles do toxic users play within and between those communities?”

Using the reddit r/gaming subreddit dataset found, we plan to investigate the below:

1. What discussion communities can be identified with the r/gaming subreddit?
2. What are the sentiment levels across all identified communities?
3. Which network roles within are associated with negative sentiments?

Success Criteria

Our research is considered successful if meaningful communities are detected, sentiment scores on all text allows for average sentiment to be calculated for various communities identified and centrality measures allow for clear differentiation of user network roles. We want to achieve a result that produces a clear understanding on how toxic users affect negatively scored average communities.

2. Data Collection & Description

Data Source

After evaluating multiple social media sources, the team found that for example YouTube was not a suitable platform to use for constructing a meaningful network graph as on YouTube, comment replies are often disconnected and confined to single videos which limits our teams ability to detect meaningful discussion communities due to the lack of an interconnected reply structure.

Upon further research, the team identified the **MADOC** (Multi-Platform Aggregated Dataset of Online Communities) which provided archived Reddit data within its repository. The **r/gaming** subreddit dataset was selected for analysis as it contains the data structure required to draw a meaningful network graph. It contained user and parent user ID for each post/comment which enables directed edges graph building and each post/comment includes textual content with only URLs extracted from the original content that is suitable for sentiment analysis.

Data Collection Method

The dataset used in this research was downloaded from MADOC with ``reddit_gaming_madoc.parquet`` file from <https://zenodo.org/records/15690964> which contains **44,274,130** rows of r/gaming data from years **2014 to 2020**.

 **DATASET SIZE:**
Total rows: 44,274,130
Total columns: 16

 **DATE RANGE:**
From: 2014-01-01 00:00:04 (Timestamp: 1388534404)
To: 2020-12-31 23:59:57 (Timestamp: 1609459197)
Duration: 2556 days

Records by year:
2014: 4,842,491 (10.9%)
2015: 4,733,256 (10.7%)
2016: 4,961,830 (11.2%)
2017: 6,255,734 (14.1%)
2018: 8,184,988 (18.5%)
2019: 7,578,095 (17.1%)
2020: 7,717,736 (17.4%)

 **INTERACTION TYPES:**
COMMENT: 41,386,090 (93.5%)
POST: 2,888,040 (6.5%)

Dataset Overview

After loading the raw dataset, we performed a sanity check on the values available before preprocessing data steps. The dataset contained of 93.5% comments and 6.5% posts, which shows that the dataset offers a dense network of reply relationships that is ideal for constructing directed network graphs. A total of 4,133,256 unique users were identified within the full dataset where on average each user contributes 10.71 interactions, but the median of interactions is only 2. This difference indicates that most users are inactive and a small subset of users instead drive most of the interactions. An examination on missing values in each column was done which revealed that `parent_id` and `parent_user_id` was missing which is expected as posts do not have such parent references. No duplicate rows were detected but 16 empty content rows of content `<NA>` text were found.

```
# Check missing content
missing_content = df['content'].isna().sum() + (df['content'] == '<NA>').sum() + (df['content'] == 'na').sum()
print(f"Actual missing content rows: {missing_content}")
```

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To focus on the latest available dataset and for computational efficiency, the team decided to restrict the analysis to the year 2020 which results in a working dataset of 7,717,736 rows.

Ethics & Limitations

Only publicly available Reddit data was used for analysis in this research. The dataset also contains anonymized IDs `user_id`, `parent_id` and `parent_user_id` which prevents any direct identification of users and no attempts to de-anonymize users were done by the team. The analysis only covers the `r/gaming` subreddit and only data in year 2020 was used due to computer constraints.

PART A – APPROACH

3. Analytical Approach

Analytical Framework: NLP + Network Integration

Our study includes Natural Language Processing (NLP) and Social Network Analysis to investigate negative sentiment within `r/gaming` subreddit for the year 2020. This approach allows us to understand our research question “Which gaming communities exhibit the most negative sentiment, and what roles do toxic users play within and between those communities?”.

The framework has 4 main components:

1. Network construction as a directed graph with users as nodes and replies/comments as edges would allow for identification of interaction patterns.
2. Community detection using Louvain algorithm with undirected graph to group users who interact together frequently as a community.
3. Sentiment analysis using VADER would categorise and score each post/comment to allow for measure of toxicity at community and individual user level.
4. Identification of network roles using various centrality measures would allow for classification and analysis of different types of users such as spreaders, targets of toxicity.

The framework would allow us to understand how toxicity works within group structures.

Research Design and Workflow

The research follows a sequential design as below:

1. Collect and preprocess the original r/gaming data collected.
2. Build a reply network where users are represented by nodes and replies/comments are represented as edges using NetworkX.
3. Detect distinct discussion communities within the network using Louvain.
4. Calculate the sentiment scores for each post/comment using VADER.
5. Identify communities with highest proportion of negative sentiment posts/replies.
6. Extract common keywords from low-sentiment score posts/comments to understand the drivers of toxicity.
7. Compute centrality measures to identify spreaders/targets/drivers of toxicity using NetworkX.
8. Analyse the overall toxicity patterns across the top 5 largest communities.

Tools

We used several packages such as Pandas, PyArrow, NetworkX, python-louvain, NLTK, Matplotlib, WordCloud, Gephi to process, analyse and visualise our research from the original raw dataset.

4. Network Analysis

Data Preprocessing

Before using the dataset for network and sentiment analysis, the team applied preprocessing steps to prepare the raw dataset downloaded. In the

`1.community\_detection.ipynb` file:

1. Column Selection and Date Filtering

```
columns_to_keep = [
    'post_id',
    'publish_date',
    'user_id',
    'parent_id',
    'parent_user_id',
    'interaction_type',
    'content'
]

df = pd.read_parquet(original_dataset_path, columns=columns_to_keep)

# Filter for posts in 2020
df['publish_date'] = pd.to_datetime(df['publish_date'], unit='s')
df = df[df['publish_date'] >= '2020-01-01']
print(f"After 2020 date filter: {len(df):,} rows")
```

We selected 7 columns from the original 16 columns that is relevant to our analysis:

'post_id', 'publish_date', 'user_id', 'parent_id', 'parent_user_id', 'interaction_type', 'content'. We then filtered the dataset for rows from year 2020.

2. Missing Content

```
missing_content_mask = [
    df['content'].isna() |
    df['content'].isin(['<NA>', 'na', 'NaN', 'None']) |
    (df['content'].astype(str).str.strip() == '')
]

missing_content_count = missing_content_mask.sum()
df = df[~missing_content_mask].copy()
```

16 rows with missing content with values of None and empty strings <NA> etc. were removed.

3. Self Replies

```
# Remove Self-Replies
self_reply_mask = (df_filtered['interaction_type'] == 'COMMENT') & (df_filtered['user_id'] == df_filtered['parent_user_id'])
# Filter out df values that are not part of self_reply_mask
df_filtered = df_filtered[~self_reply_mask]
print(f"Removed {self_reply_mask.sum():,} self-reply rows")
```

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We removed self replies where 'user_id' equals 'parent_user_id' as they do not contribute meaningful social interactions as it would introduce self-loops in the network graph.

4. Comments all have 'parent_id' and 'parent_user_id'

```
posts_mask = (df['interaction_type'] == 'POST')
comments_mask = (df['interaction_type'] == 'COMMENT')
valid_comments_mask = comments_mask & df['parent_user_id'].notna() & df['parent_id'].notna()
df_filtered = df[posts_mask | valid_comments_mask].copy()
comments_removed = len(df) - len(df_filtered)
```

We checked that the dataset comments type row had valid 'parent_id' and 'parent_user_id' attached as it is an important field to construct directed reply edges between users.

These preprocessing steps resulted in a final **3,374,714** rows that would be used for further analysis. Saved **3,374,714 rows** to data/filtered_gaming_2020.parquet

Network Construction

The cleaned data for year 2020 was used to construct a weighted directed reply network using NetworkX where Nodes represent unique users who posted to r/gaming within the timeframe, with directed Edges representing reply interactions, where an edge from User A to User B would indicate that User A replied to User B. Each edge is then weighted for the frequency of replies between the pair of users.

```
reply_graph = nx.DiGraph()
```

All user IDs available was added as Nodes in the reply network.

```
all_users = pd.concat([df['user_id'], df['parent_user_id']].dropna()).unique()
for user in all_users:
    reply_graph.add_node(user)
```

For each comment, an edge was added from commenting user to parent user and if multiple replies were identified, the edge weight was incremented by 1 each time.

```
for _, row in comments.iterrows():
    # Check that row is a comment
    if row['user_id'] != row['parent_user_id']:
        if reply_graph.has_edge(row['user_id'], row['parent_user_id']):
            reply_graph[row['user_id']][row['parent_user_id']]['weight'] += 1
        else:
            reply_graph.add_edge(row['user_id'], row['parent_user_id'], weight=1)
    edge_count += 1
```

The resulting graph constructed was saved [[data/gaming_reply_graph.graphml](#)] for reuse in subsequent analysis.

Network Overview

Metric	Value
Total nodes (users)	979,631
Total edges (unique reply pairs)	2,857,838
Network density	0.000003
Largest connected component	880,447 nodes (89.9% of all nodes)
Maximum edge weight	46 (46 replies from one user to another)

The low network density (0.000003) is consistent with social network theory for large online communities as most users interact with only a small percentage of total users. The 89.9% largest connected component is notable as it indicates that in r/gaming discussion communities operate in conjunction with one another as a large cohesive community rather than a collection of isolated groups. This high connectivity suggests

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that it is possible for toxicity to spread easily and impact users. A subset of 71,324 nodes (7.3% of total) appears as reply targets but have no comment data associated as their original posts/comments were made prior to the 2020 timeframe but are still receiving replies. These nodes would be retained in the network to preserve graph structure but will be excluded from sentiment analysis due to lack of textual content, which may introduce uncertainty into top centrality rankings.

Community Detection

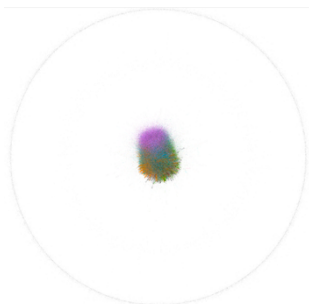
To identify community groups within the r/gaming reply network constructed above, we applied Louvain algorithm. As Louvain requires undirected graph, we converted the directed reply graph into an undirected reply graph disregarding whoever initiated the reply.

```
undirected_reply_graph = nx.to_undirected(reply_graph)
```

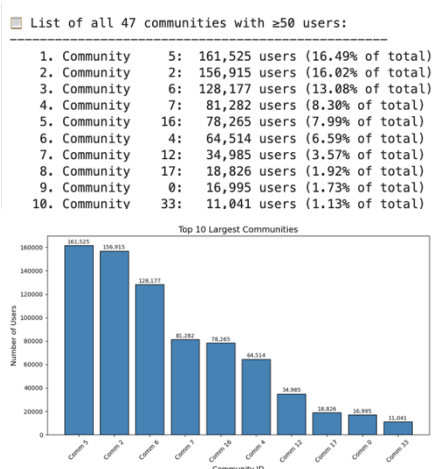
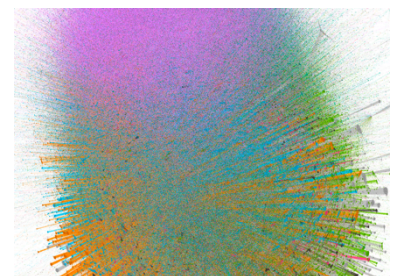
After running the Louvain algorithm, it detected 97,299 communities with a modularity score of 0.4075 from the original 979,631 nodes and 2,857,838 edges. The modularity score indicates that the community identified is more “tightly knit” [1] meaning users within the community interact more frequently within the community than with other communities.

```
louvain_partition = community.best_partition(undirected_reply_graph)
modularity = community.modularity(louvain_partition, undirected_reply_graph)
```

We then saved the community identified as a node attribute ‘community’ and saved the updated reply graph [[data/gaming_graph_with_communities.graphml](#)] to visualise in Gephi.



The full Gephi reply network appears as a dense central cluster of nodes, with each community represented as different colors which confirms the presence of a largely connected network graph.



Using the insights available from the Louvain algorithm, we find that the largest community 5 contained 161,525 users with only a total of 47 communities containing more than 50 users in each. This meant that majority of the identified communities are small and isolated. We find that the top 5 communities identified (5,2,6,7,16) account for over 60% of all users in the network therefore further confirming that discussion is concentrated within a small number of communities.

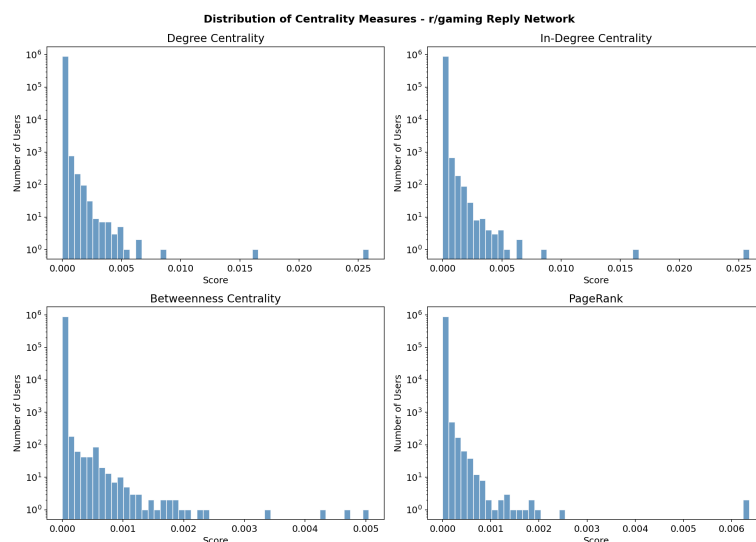
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Centrality measures

Four centrality measures were computed on the network to establish user-roles and determine user-influence.

Measure	Purpose
Degree Centrality	The proportion of all users that each user has a connection to
In-Degree Centrality	Which users receive the most replies
Betweenness Centrality	Which users bridge communities (approximated, k=200)
PageRank	Influence weighted by the importance of connecting users rather than only connection count

All the measures follow a heavily right-skewed power law distribution, again consistent with social network theory. A small minority of highly active users dominate while the vast majority have near-zero centrality scores (Figure 4).



There is an overlap between the top users by degree and in-degree centrality, which implies that users who attract many replies rather than users who initiate many replies are the most connected.

Figure 4

5. NLP Analysis

Data Preprocessing

Two distinct pipelines were applied: one for topic modelling and another for sentiment analysis (this depended on the downstream task). For topic modelling, the pipeline uses an aggressive heavy cleaning, while the pipeline for sentiment analysis uses a lighter cleaning (this was on purpose). All text is converted to lowercase, and emojis are also removed just for topic modelling. For both pipelines, the URL is removed using regex and if a user's mentions (@user). Additionally, the hashtag # symbol is eliminated,

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while we have kept the word. Using the **NLTK English stop words**, a very detailed list of stop words was created with those lists:

- Internet informal expression and any filler terms, for example, (omg, lol, Ugh, Smh, etc.)
- Character that includes punctuations (Unicode forms, curly quotes, em-dashes, etc.)
- Verbs that provide limited topical values that are related to sentiment, such as (feel, know, think, etc.)

Plus, we have made sure to remove tokens that contain digits; this is only applied to topic modelling. In contrast, whitespace was normalised for sentiment analysis. intentionally punctuation, casing and informal language were kept because they tend to have affective cues and emotions that are important to sentiment models.

Tokenisation & lemmatisation

NLTK's Tweet Tokenisers is used for tokenisation on topic analysis. This is designed and made to work primarily for a social media text, which can handle complex slang and informal writing better than a standard tokenisers. After tokenisation is done, each token is whitespace-stripped. To reduce words that are inflected from their dictionary base, lemmatisation is applied using WordNetleeatizer. Example of words after this is applied ("running" > "run" >, "Companies" > "company"). Again, after lemmatisation is done, the stop word filter is applied; it works by catching base forms that were not caught beforehand. Pipeline of sentiment does not lemmatise or tokenise; it just returns a clear string.

Data transformation

Different outputs from the two pipelines can be observed, with different structures reflecting their intended uses. Sentiment output is a cleaned string that is ready for a **lexicon or sentiment classifier**; on the other hand, Topic output is a ready list of tokens for topic modelling like **LDA** or word frequency count. To prevent errors on bad data, both functions are made to return an empty result if there is a missing or non-string input.

Sentiment analysis

Sentiment Analysis was conducted on a large dataset of comments (Reddit database), to be exact, **3,192,856** comments, with **3,190,746** valid entries, which equated to 99.93 per cent after preprocessing. But the primary focus of this analysis depended on the top 5 largest communities (**2,431,483 comments / 76.2 per cent of the dataset**).

VADER Sentiment analysis method was used, which is predominantly designed to work for social media text. It works by producing a compound score from -1 (negative) to + 1

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(positive). As mentioned in the data preparation section, preprocessing for sentiment removed URL and mentions (@users), but the structure of the sentence was kept. A low sentiment threshold is used, defined as a score of < -0.3 (for downstream analysis, it is great at locating highly negative comments). VADER uses a lexicon-based approach for sentiment scoring, also polarity score is given for each word (positive or negative). This works by context rules (Negation, intensifiers, capitalisation, and punctuation emphasis). Each comment is assigned a compound sentiment score, which is calculated at the individual comment level. VADER was chosen for a couple of reasons: it handles slang, emojis, and informal writing; also, it does not require training data (which can take time), and it is a pre-trained model. Later, sentiment values are aggregated at the community level for comparative analysis.

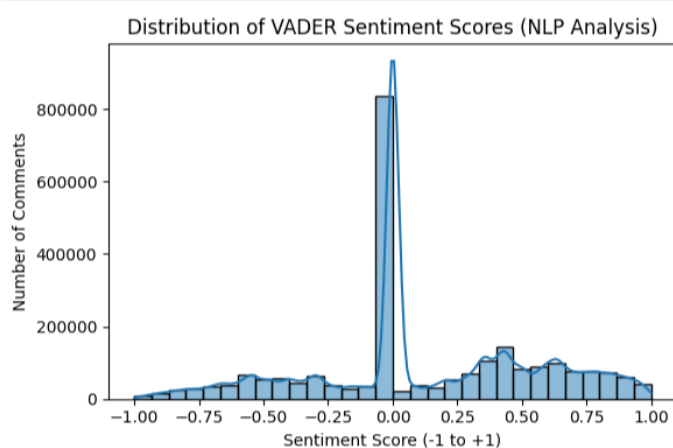


Figure 5:

This histogram shows the distribution of the VADER sentiment score assigned to Reddit comments (after preprocessing and sentiment scoring)

Topic modelling

To discover hidden patterns and themes within Reddit post topic modelling is used. LDA (Latent Dirichlet Allocation) is selected as a modelling approach, due to the size of the database; manual analysis was not used. So, automated topic extraction was required to process and analyse the text efficiently. As mentioned previously, **3,192,856** reddit post is collocated for analysis. Social media posts (text) generally tend to be unstructured; therefore, preprocessing was a necessary step before any topic modelling could be applied. This ensured that the data was suitable for extracting any meaningful patterns.

To support modelling, the raw text was transformed into structured word-level representations. Improving model accuracy is important for the success of topic modelling, because of this only relevant and meaningful tokens were retained for model accuracy. The post that has no usable tokens after cleaning was removed to protect the quality of data and consistency during topic extraction.

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Metric	Value
Total Rows	3,192,856
Valid Rows (Topic modelling)	3,146,680
Excluded Rows	46,176
Retention rate	98.55%

Model setup & parameters.

The table below is a summary of the model setup and parameters used in the NLP analysis.

Component	Setting	Description
Topic Model	LDA (Latent Dirichlet Allocation)	Probabilistic model used to extract latent topics from text
Input Representation	Bag-of-Words	Converts documents into word frequency vectors
Number of Topics (k)	Predefined value	Controls the number of topics generated by the model
Training Passes	Multiple iterations	Improves model convergence and topic stability
Random State	Fixed seed	Ensures reproducible results across runs
Sentiment Model	VADER	Lexicon-based sentiment analysis model for social media text
Sentiment Score	Compound score (-1 to +1)	Measures overall polarity of text
Classification Method	Threshold-based rules	Assigns positive, neutral, or negative labels based on score

Key patterns

1. Pretty much all top communities show a proportionally similar low sentiment post around (15 % to 17 %)
2. There is a shared vocabulary and similar language use across the community's different user groups.
3. Informal language use tends to dominate, including slang, abbreviations, and emotionally expressive words.
4. Memes and short reaction-style posts tend to contribute overall to the dataset of text structure.
5. Negative sentiment post form clear lexical clusters containing words such as ("hate" , "shit" , "kill" and "suck" etc)

PART B – RESULTS

6. Combined Results

Sentiment results

Sentiment scores were aggregated at the community level to compare emotional patterns across the top five largest communities. From what is observed, average sentiment values across communities ranged from **(0.0939 to 0.01551)**, which shows a mild positive tone. No extreme polarity was observed; all communities show a similar sentiment distribution. Among all groups, community 2 recorded the lowest average sentiment at **0.0939**. In addition, Community 2 also had the highest proportion of low sentiment posts (**17.17 %**). Other communities showed a relatively low sentiment proportion, ranging between **15.50% and 16.28 %**. Across all communities, negative sentiment posts were consistently present. Complaint-related or frustration-based language made up most of the low sentiment content. Sentiment variation between communities was small, which proves that emotional behaviour patterns were largely similar.

This heatmap (figure 6) compares sentiment and activity across the top five communities. Community 5 happens to have the highest post volume and the most positive overall sentiment. Community 2 shows the highest low sentiment rate and lowest sentiment. Sentiment is not the main source of variation; it is activity level. A large variation in total posts exists among all communities.

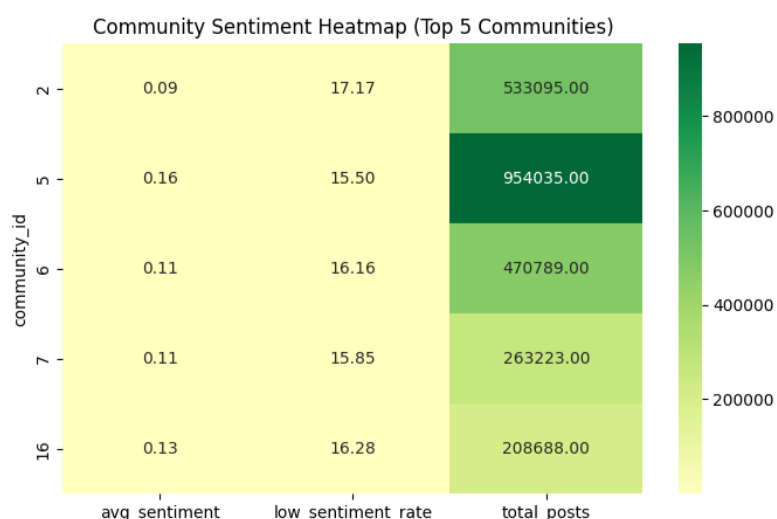


Figure 6

Unlike the heatmap (figure 7), this box plot focuses on the internal distribution of sentiment scores within each of top five communities. It displays the distribution of

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sentiment scores within each community. Majority of sentiment values cluster at above zero, there is a similar internal structure across communities. This is a comparable spread of values. It also highlights the consistent presence of negative outliers in all the communities. Furthermore, it reveals that positive sentiment is more frequent but negative extremes still exist. Despite the difference that is shown in activity levels the structure is similar around community distributions.

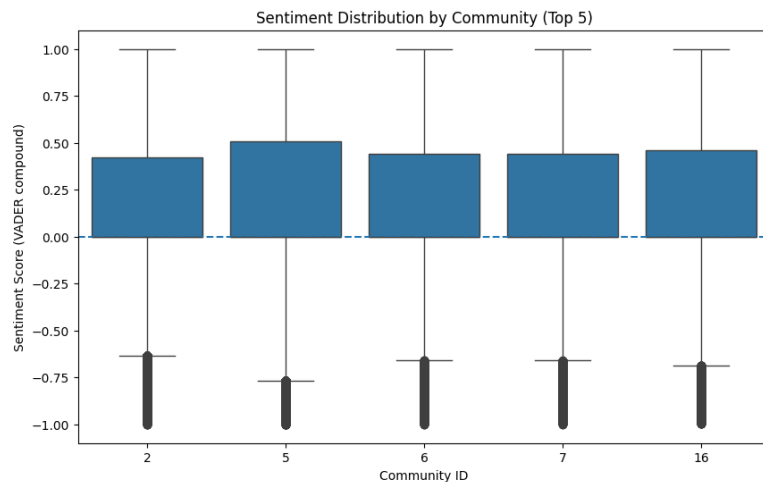


Figure 7

Topic modelling results

Topic modelling produced structured outputs for 3,146,880 valid posts. Each post is represented as a set of (tokens) which is set of extracted keywords. Output observed consist of (word level token group per post). Some post contained single -token outputs, while other post had multi-token outputs. Across the dataset token outputs show frequent redundancy of core term. The most frequently appearing token groups are:

- Game Related terms: (game, play, console, xbox)
- Negative or opinion-based terms: (bad, hate, fu*k, damn)
- General discussion terms: (people, time, make, thing)

The dataset output is totally made of (tokenised post -level results). Outputs of the keyboard do vary, based on size and complexity between posts.

Looking at word cloud also known as tag cloud or weighted list (figure 8). Eight topics (community) show distinct keyword grouping, the evidence show that gaming related terms dominate across all topics. Each topic highlights different frequent words and clusters. Some have hardware term languages (xbox, console , pc) while others have sentiment word like (good, bad , love , great).

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Figure 8

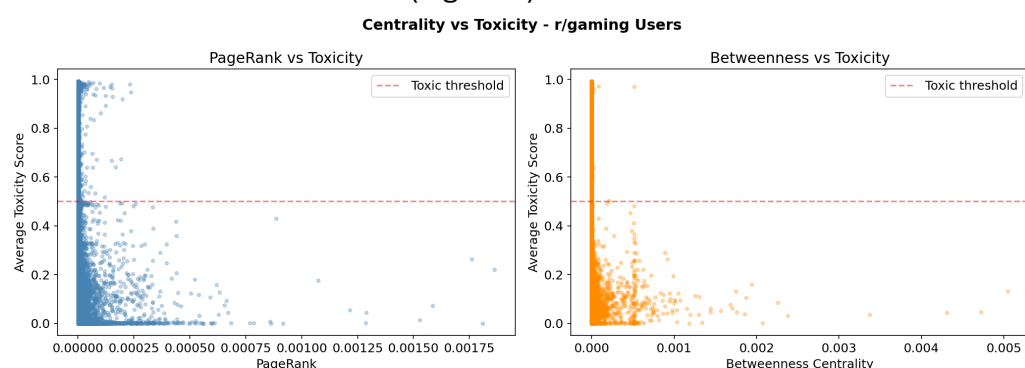
Network results

Centrality and Toxicity

A classification was applied to each user based on their ToxiGen scores (a measure of their toxicity). Users were classified as toxic if their ToxiGen average was above 0.3. This was lowered from the original threshold of 0.5 due to a surprising lack of toxicity overall, as evidenced by the extreme right skew in Figure 1, and the classification of only 1% of the total userbase as toxic using a 0.5 threshold. 10.3% (84,451) of the users were classified as toxic, out of 819,324 total users. Classification of toxic users in this analysis was based on ToxiGen scores (0.3 threshold), whereas VADER compound scores were used for the analysis in section 7. While similar, these scores should not be considered interchangeable. Correlation between toxicity and sentiment was computed on the full userbase and indicated moderate correlation.

Measure	Non-Toxic Users	Toxic Users
Degree Centrality	0.000006	0.000004
In-Degree Centrality	0.000002	0.000002
Betweenness Centrality	3.83e-07	8.35e-08
PageRank	8.91e-07	7.66e-07

Compared to toxic users, non-toxic users show consistently higher centrality scores across each of the measures (Figure 9).



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Figure 9

This is supported by the network visualisation, with the most active and central user nodes being predominantly non-toxic (Figure 10). Across all PageRank quartiles (0.084-0.092) toxicity scores are mostly uniform, suggesting a lack of correlation between level of influence and toxicity.

Though toxic users overall have lower PageRank and degree centrality, they show similar in-degree centrality to non-toxic users, suggesting that toxic users attract more replies than expected in relation to their connectivity. This is consistent with provocative content generating reactive engagement without having significant actual influence.

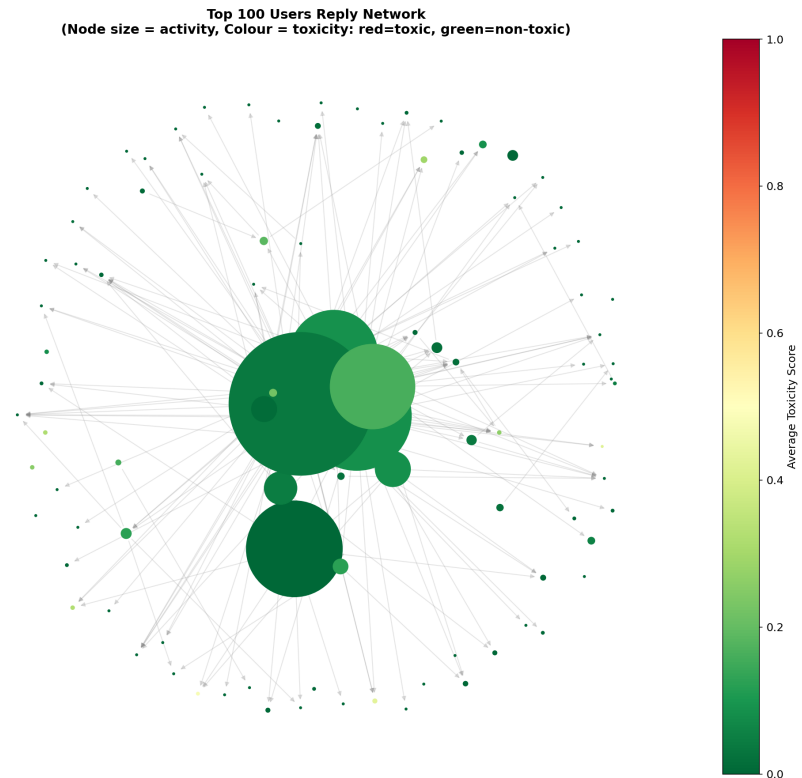


Figure 10

Community Centrality and Toxicity Analysis

Community labels from Louvain community detection were used to conduct centrality and toxicity analysis within communities. 810,816 users of 819,324 (98.9%) with centrality data were matched to community assignments.

Community	Users	Avg Toxicity	Toxic User %
Degree Community 5 (largest)	161,525	0.067	6.78%
Community 2	156,915	0.104	12.05%
Community 6	128,177	0.095	10.88%
Community 7	81,282	0.097	11.29%
Community 29 (least toxic)	1,655	0.042	5.00%
Community 75 (most toxic)	1,725	0.126	14.66%

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Community 2 had one of the highest average toxicity scores among the large communities (0.104), intuitively suggesting the most active discussion group in r/gaming has the most toxic users overall.

The relationship between community toxic user percentage and average PageRank shows no meaningful pattern (Figure 11), consistent with the finding that toxicity and influence are largely independent. Large communities cluster around 10-12% toxic users with similar PageRank values, suggesting the major communities are structurally similar.

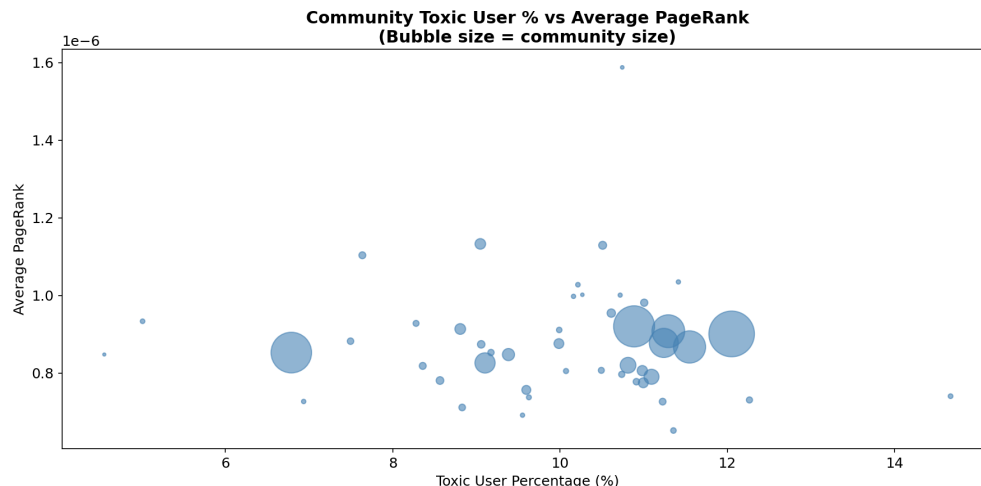


Figure 11

In communities 25 and 6 are the highest PageRank toxic users (0.48 and 0.43), suggesting that within these communities, toxic users have actual influence. Compared to the largest communities, where toxicity is not concentrated in specific users but across the community as a whole, the impact of toxic users can be seen more clearly, and may be amplified, in smaller communities. Community 75's high toxicity score should be scrutinised due to its small size, which may cause it to be more susceptible to influence from individual toxic users.

PART C – DISCUSSION

7. Discussion

NLP Interpretation

The sentiment analysis revealed that most comments in their respective community are positive with the average sentiment score being positive ranging from 0.0939 onwards. This suggests that while low sentiment (<0.3 sentiment score) comments do exist ranging at approximately 16% of all comments within each community, they are not making up majority of the discussions. The topic modelling analysis indicate that recurrent keyword cluster are present across all communities, but much of strong presence of gaming related topics and terms are (play, console, xbox, game). There is emotional language that appear frequently too like (bad, damn, Fu*k, hate) also general

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discussion words such as (things, make, time, people) overall from what we can observe similar keyboard patten can be seen across all communities.

Network Interpretation

Toxic users in r/gaming do not significantly influence r/gaming at the network level. The users with higher activity, connectivity, degree centrality and PageRank are predominantly non-toxic, suggesting r/gaming is made up of overall positive attitudes and constructive viewpoints.

Whilst toxic users do not highly influence the network, they do attract comparable replies to non-toxic users despite on-average lower connectivity. This is aligned with the dynamic of online communities in which provocative content generates engagement in reaction without true influence or connection. Toxicity is amplified through the reaction of other users, resulting in seemingly higher influence.

This is true at the network level, but within communities there are toxic users with high centrality and influence, suggesting that toxicity does not prevent connectivity between users and is not defined by lack of centrality.

Non-toxic users are more likely to bridge between communities, as evidenced by the betweenness centrality of non-toxic users ($3.83e-07$) compared to toxic users ($8.35e-08$). This is contrast to the expectation of negativity spreading across the network, rather remaining concentrated in the discussion groups it originates in.

The findings suggest that toxicity within r/gaming is a reactive phenomenon, rather than influential itself. Toxicity attracts replies but not connection in the larger network, but in smaller communities, toxic users can still drive engagement and have real influence.

Integrated Insight

Integrating both network and sentiment interpretation reveals how toxicity operates within and between communities. Across the network, toxic users receive replies of a similar volume to non-toxic users (same in-degree) but have a lower PageRank than non-toxic users which suggests that though toxic users attract engagement and replies they do not achieve significant network influence. Cross community discussions are found to be led by non-toxic users with their higher betweenness centrality scores which suggests that toxicity remains concentrated within its own community not affecting other communities.

8. Conclusion

Success Criteria Evaluation

Our research is considered successful as we successfully identified meaningful discussion communities, individual sentiment scoring for all posts/comments which allowed for average sentiment per community to be scored and centrality measures differentiated user network roles which allowed us to understand the functionality of all users (toxic and non-toxic) on negatively scored sentiment communities.

Research Question Answered

Our research question “Which gaming communities exhibit the most negative sentiment, and what roles do toxic users play within and between those communities?” has been answered through our sentiment, network community and centrality analysis.

“Which gaming communities exhibit the most negative sentiment”:

Community 141 with 2,704 users has the lowest average sentiment recorded at 0.04.

“What roles do toxic users play within communities”:

Toxic users attract engagement and replies but they do not achieve significant network influence or serve as bridges between different communities. The replies that toxic users gather are usually reactive replies that do not translate to higher influence or central discussions.

“What roles do toxic users play between communities”:

Toxic users remain discussions within their own community and do not lead cross-community discussions instead non-toxic users facilitate more cross community discussions as can be seen from their higher betweenness score.

Final Takeaways

Overall, toxicity in r/gaming generates engagement but does not translate into network influence or bridging capacity. Our findings show that communities respond to toxic users but do not elevate them to central positions. This suggests that while toxicity exists, the r/gaming community does not reward or amplify negativity at the network level which means that toxic users do not affect or influence the spread of toxicity within/between communities.

9. Limitations & Future Work

Data limitations

We did the analysis on r/gaming using an alternative source MADOC data due to the unavailability of current Reddit API access. Our filtering of dataset to only include 2020 data resulted in 7.3% of nodes being retained as network nodes but have missing content in the dataset due to the content being published before our 2020 filtered range.

Method limitations

VADER sentiment analysis may unintentionally score sentiment scores wrongly for some text due to its inability to detect sarcasm or gaming puns.

Limitations of Centrality Analysis

- 0.3 toxicity threshold is an analytical choice. Results may differ with alternative thresholds.
- Analysis treats all reply interactions equally regardless of content.

10. References

- [1] A. Mishra, "Demystifying Louvain's Algorithm and Its implementation in GPU," [Online]. Available: <https://medium.com/walmartglobaltech/demystifying-louvains-algorithm-and-its-implementation-in-gpu-9a07cdd3b010>.